MILITARY SPECIFICATION

SYSTEM OPERATING UNIT LA-205( )
DETECTOR, LIGHT INTENSITY LA-210( )

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for System Operating Unit LA-205( ), Detector, Light Intensity LA-210( ) used for daytime photography in an airborne reconnaissance camera control system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods Of.
MIL-M-13231	- Marking Of Electronic Items.
MIL-F-14072	- Finishes For Ground Electronic Equipment.
MIL-E-55585	- Electronics Equipment And Parts, Packaging Of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-252	- Wired Equipment, Classification Of Visual And Mechanical Defects.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6760

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MIL-STD-454 - Standard General Requirements For Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirement For The Control Of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of.

MIL-STD-704 - Aircraft Electric Power Characteristics.
MIL-STD-726 - Packaging Requirements Code.
MIL-STD-810 - Environmental Test Methods.

2.1.2 Other Government documents. The following Government documents form a part of this specification to the extent specified herein:

DRAWINGS

ELECTRONICS COMMAND

SC-DL-445503 - Automatic Light Detector.
SM-A-445980 - Final Operational Test Procedures, Automatic Light Detector.
SM-C-445987 - Output Graph.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The detector utilizes a photoelectric cell and an amplifier to produce a signal proportional to the average terrain brightness below the airborne vehicle. This signal is used to set the camera shutter speed and lens aperture for optimum exposure according to existing light conditions.

3.2 First article. The quantity of samples specified in the contract shall be subjected to first article inspection (see 4.3 and 6.2b).

3.3 Material. The material shall be as specified herein and as shown on the applicable drawings. However, when a definite material is not specified, a material shall be used which will enable the equipment to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirement of this specification.

3.4 Design.

3.4.1 Electrical.

3.4.1.1 Input, voltage and frequency. The input to the detector shall be in accordance with the following power sources with characteristics as defined in MIL-STD-704 having ranges as specified below:

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a. AC power. 3 phase, 115 volts; operating voltage range: 108 to 118 volts; operating frequency range: 380 to 420 Hertz.
b. DC power. 28 volts; operating voltage range: 25 to 29 volts.

3.4.2 Construction. The detector shall be constructed in accordance with Dwg. SC-DL-445503 and as specified herein.

3.4.3 Operation. During operation, the detector output voltage shall fall within the limits of Dwg. SM-C-445987.

3.4.4 Acceleration. During operation, the detector output voltage shall fall within the limits of Dwg. SM-C-445987. The detector shall withstand acceleration of 4 times the force of gravity (g's) at operating and 6 g's at non-operating conditions.

3.5 Environmental conditions. The detector shall withstand individual environmental conditions specified herein, or combinations thereof, without damage, performance degradation, or malfunction when tested as specified in 4.6.3.

3.5.1 Temperature.

3.5.1.1 Non-operating. The non-operating temperature range shall be -60°F to +160°F (-51°C to +71°C). During operation, the detector output voltage shall fall within the limits of Dwg. SM-C-445987.

3.5.1.2 Operating. The operating temperature range shall be -25°F to +105°F (-32°C to +41°C). During operation, the detector output voltage shall fall within the limits of Dwg. SM-C-445987.

3.5.2 Relative humidity. The detector shall withstand the effect of relative humidity up to 100 percent during operating and non-operating conditions and shall meet the requirements of 3.4.3.

3.5.3 Dust. During operation, the detector output voltage shall fall within the limits of Dwg. SM-C-445987.

3.5.4 Altitude. The upper operational altitude limits shall be 20,000 feet (6.1km). During operation, the detector output voltage shall fall within the limits of Dwg. SM-C-445987.

3.5.5 Vibration. During operation, the detector output voltage shall fall within the limits of Dwg. SM-C-445987.

3.5.6 Shock. During operation, the detector output voltage shall fall within the limits of Dwg. SM-C-445987. The detector shall withstand a maximum of 7.5 g's.

3.5.7 Fungus. After the fungus test of 4.6.3.6, the detector shall be visually examined, using a 10-power magnifier. The detector shall show not more than six minute unrelated spots, each no greater than 0.015 square inch (9.7sq mm) in area, of sparse microbial growth as evidenced by growth colonization (which includes branching and sporulation) on or within each cubic foot, or fraction thereof, of detector assembly volume. Isolated instances of partial tubular germination shall not be included in this evaluation (see 6.4).

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3.6 Electromagnetic interference characteristics. The detector shall meet the electromagnetic interference requirements specified in MIL-STD-461, table 1-11, class C3.

3.7 Preconditioning. The detector shall be capable of meeting the performance requirement herein, without subsequent processing, after subjection to 25 hours continuous electrical operation.

3.8 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.9 Finish. The detector shall be finished in accordance with MIL-F-14072 and the equipment drawings.

3.10 Marking. Marking shall conform to MIL-M-13231.

3.10.1 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the detector.

3.10.2 Serial numbers. The detector shall be serial numbered.

3.11 Workmanship. The detector shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.6.8).

4.3 First article.

4.3.1 First article units. When specified in the contract, the contractor shall furnish first article units of the detector (see 6.3).

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4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, the preconditioning specified in 4.6.5, and the inspections specified for group A, group B, and group C. The inspection shall be performed in the following order: 4.6.5, 4.4, group A and group B for all units. and designated units within group C in accordance with 4.5.3.1. After completion of group C environmental tests, conforming units shall be reinspected and shall pass all group A inspections. Failure of any inspection shall be cause for rejection.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection requirements:

Item	Where required
Material	3.3
Finish	3.9
Marking	3.10
Workmanship	3.11

4.5 Quality conformance inspection. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.3.3, and 4.6.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit shall be subjected to the inspection specified in table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined, or as specified herein, shall be in accordance with MIL-STD-105.

TABLE I. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (Percent defective)	
			Major	Minor
Visual and mechanical	3.11	4.6.7	1.0	4.0
Electrical	3.4.1	4.6.1	1.0	1/

1/ All electrical defects are major.

4.5.1.1 Group A lot sampling. Sampling of group A lots shall be in accordance with MIL-STD-105, inspection level II. Acceptable quality levels (AQL) shall be specified in table I.

4.5.1.2 Order of inspection within group A. Group A inspection shall be performed in any order satisfactory to the Government.

4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection. This inspection shall consist of inspection specified in table II and shall conform to the inspection level S-4 of MIL-STD-105.

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TABLE II. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (Percent defective)
Interchange-ability	3.8	4.6.6	6.5

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspection specified in table III.

4.5.3.1 Sampling for group C inspection. Two sample units shall be selected at random for every 50 units, or fraction thereof, produced. The first sample selected shall be at the start of the contract from the first quality conformance inspection production lot.

TABLE III. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Test Sequence	
			Unit 1	Unit 2
Acceleration (Operating)	3.4.4	4.6.2.1	3	
Acceleration (Non-operating)	3.4.4	4.6.2.2	4	
Temperature (Non-operating)	3.5.1.1	4.6.3.1.1	1	
Temperature (Operating)	3.5.1.2	4.6.3.1.2	1	
Relative humidity	3.5.2	4.6.3.2	2	
Dust	3.5.3	4.6.3.3		1
Altitude	3.5.4	4.6.3.1	1	
Vibration	3.5.5	4.6.3.4		2
Shock	3.5.6	4.6.3.5		3
Fungus 1/	3.5.7	4.6.3.6		4
Electromagnetic interference	3.6	4.6.4	5	

1/ The detector shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed as specified in table III.

4.5.3.3 Group C failure. Action required relative to group C failure shall be as specified in the contract (see 6.2c).

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4.5.3.4 Reinspection of conforming group B and C sample units. Unless otherwise specified, sample units which have been subjected to and pased both group B and group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

4.6 Methods of inspection.

4.6.1 Electrical inspection. The detector shall be tested in accordance with the procedures specified on Dwg. SM-A-445980. Nonconformance to 3.4.1 shall constitute failure of this test.

4.6.2 Acceleration.

4.6.2.1 Equipment operating. The detector shall be tested in accordance with method 513.2, procedure II of MIL-STD-810. The test level shall be 4 g's.

4.6.2.2 Equipment non-operating. The detector shall be tested in accordance with method 513.2, procedure I of MIL-STD-810. The test level shall be 6 g's.

4.6.2.3 Failure criteria. Nonconformance to 3.4.4 shall constitute failure of these tests.

4.6.3 Environmental.

4.6.3.1 Temperature and altitude.

4.6.3.1.1 Non-operating. The detector shall be tested in accordance with method 504.1, procedure I, table 504.1-II, category 4 of MIL-STD-810. The low temperature limit shall be -60°F (-51°C), and high temperature limit shall be 160°F (71°C). The maximum altitude shall be 20,000 feet (6.1km).

4.6.3.1.2 Operating. The detector shall be tested in accordance with method 504.1, procedure I, table 504.1-II, category 4 of MIL-STD-810. The low temperature limit shall be -25°F (-32°C), the high temperature limit shall be 105°F (41°C) and the maximum altitude shall be 20,000 feet (6.1km).

4.6.3.2 Relative humidity. The detector shall be tested in accordance with method 507.1, procedure I of MIL-STD-810, with relative humidity up to 100 percent including condensation.

4.6.3.3 Dust. The detector shall be tested in accordance with method 510.1, procedure I of MIL-STD-810, except that the second 6-hour test at 63°C (145°F) shall be performed immediately after reaching stabilization in step 2.

4.6.3.4 Vibration. The detector shall be tested under ambient conditions in accordance with method 514.2, procedure I, table 514.2-III and figure 514.2-3, part I (curve M) of MIL-STD-810.

4.6.3.5 Shock. The detector shall be tested in accordance with method 516.2, procedure I of MIL-STD-810 to determine structural integrity of the equipment under conditions of shock. The shock pulse shall be in accordance with figure 516.2-2 for flight vehicle equipment.

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4.6.3.6 Fungus. The detector shall be tested in accordance with method 508.1, procedure I of MIL-STD-810, except the test shall be 14 days. At the end of the incubation, the test item shall be removed from the test chamber, returned to ambient condition and visually inspected to determine that the detector has satisfactorily withstood conditions of the test, followed by procedures specified in 4.6.1.

4.6.3.7 Failure criteria. Nonconformance to 3.5 shall constitute failure of these tests.

4.6.4 Electromagnetic interference characteristics. The detector shall be tested for electromagnetic interference characteristics in accordance with MIL-STD-462. Nonconformance to 3.6 shall constitute failure of this test.

4.6.5 Preconditioning. Each detector when tested as specified shall operate not less than 25 hours, the last 12 hours shall be failure-free. Operation shall consist of applying various simulated light levels which span the operational range and monitoring the output signals. The test level temperature, temperature cycling, vibration, and equipment "On-Off" cycling shall be as follows:

a. Temperature. -32°C to +41°C (-25°F to +105°F).
b. Temperature cycling. Temperature cycling shall have time to stabilize at low temperature followed by time to stabilize at the high temperature, plus 2 hours.
c. Vibration. 2.2 g's ±10 percent peak acceleration value at any non-resonant frequency between 20 and 60 cycles measured at the mounting points on the equipment. The duration of vibration shall be not less than 10 minutes during each hour of operating time.
d. Equipment "on-off" cycling. Equipment shall be "off" during the heating cycle.

NOTE: Input voltage shall be the nominal voltage specified, +5 or -2 percent. Nonconformance to 3.7 shall constitute failure of this test.

4.6.6 Dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.8. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.
b. Dimensions of cavities, when such dimensions affect insertion of items.
c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

4.6.7 Visual and mechanical. The detector shall be examined for the defects listed in MIL-STD-252. Nonconformance to 3.11 shall constitute failure of this test.

4.6.8 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

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4.6.8.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.6.8.2 Quality conformance inspection of packaging.

4.6.8.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.6.8.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.6.8.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table IV, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

Table IV. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 1Ø-1-1-ØØ-00-NU-X-B7-A-ED-A.

5.1.2 Level B. 1Ø-1-1-ØØ-00-NU-X-B7-A-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

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6.1 Intended use. The detector is part of Photographic surveillance system, airborne KS-61( ), mounted in an airborne vehicle. It shall produce an AC signal proportional to the logarithm of terrain brightness.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval, and quantity of first article units specified in the contract (see 3.2).
c. Group C failure action (see 4.5.3.3).
d. Level A or B preservation and packing (see section 5).
e. When first article inspection rough handling tests are not required.
f. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article model. The first article should consist of the quantity of first article units specified in the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of first article.

6.4 Definitions.

6.4.1 Branching. Branching is a connnected arrangement of filament (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.4.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitues abundant growth.

6.4.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.4.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.4.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: LA-210W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

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6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ER

Preparing activity:
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(Project 6760-A581)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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